

(2 Hours)

[Total Marks: 60]

- N. B.: (1) All questions are compulsory.
(2) Make suitable assumptions wherever necessary and state the assumptions made.
(3) Answers to the same question must be written together.
(4) Numbers to the right indicate marks.
(5) Draw neat labeled diagrams wherever necessary.
(6) Use of Non-programmable calculators is allowed.

1. **Attempt any two of the following:** 12
 - a. List the various characteristics of Big Data.
 - b. Summarize the differences between ACID and BASE transactions.
 - c. Explain the MongoDB Design Philosophy in detail.
 - d. Describe the characteristics that differentiate the various categories of NoSQL databases.

 2. **Attempt any two of the following:** 12
 - a. What are the core processes and tools included in the MongoDB package?
 - b. Describe the roles of different secondaries in a MongoDB Replica Set.
 - c. Define a capped collection in MongoDB.
 - d. List the conditional operators available in MongoDB.

 3. **Attempt any two of the following:** 12
 - a. Explain the key features of the WiredTiger Storage Engine and its advantages over other storage engines.
 - b. Describe the limitations of MongoDB and how they may affect its use in certain scenarios.
 - c. What tips can help decide between sharding and replication?
 - d. Define GridFS in MongoDB.

 4. **Attempt any two of the following:** 12
 - a. Analyze the advantages and disadvantages of using TimesTen compared to traditional disk-based databases.
 - b. Describe the architecture of Oracle 12c In-Memory Database and its significance.
 - c. List the features supported by jQuery.
 - d. What is an Event? Explain with syntax fadeIn() and fadeOut() jQuery methods.

 5. **Attempt any two of the following:** 12
 - a. List the different JSON Data types and their significance..
 - b. Describe the structure and rules of JSON grammar.
 - c. How would you use the stringify method in a JavaScript application? Provide an example.
 - d. Describe how JSONP works and its primary use case.
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